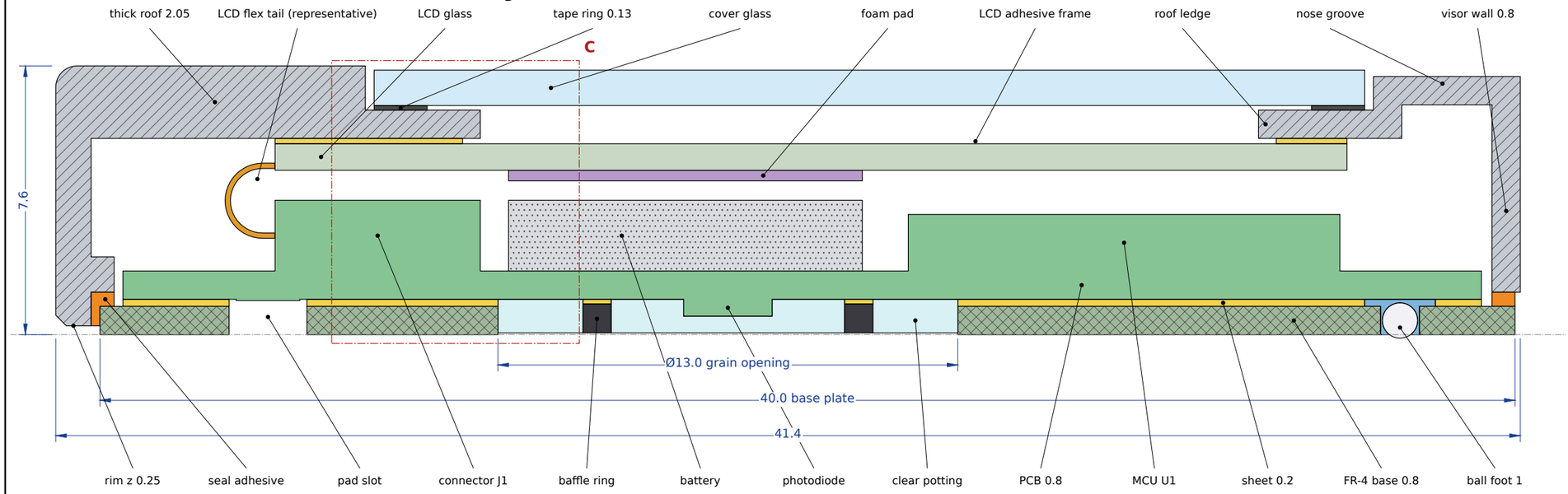
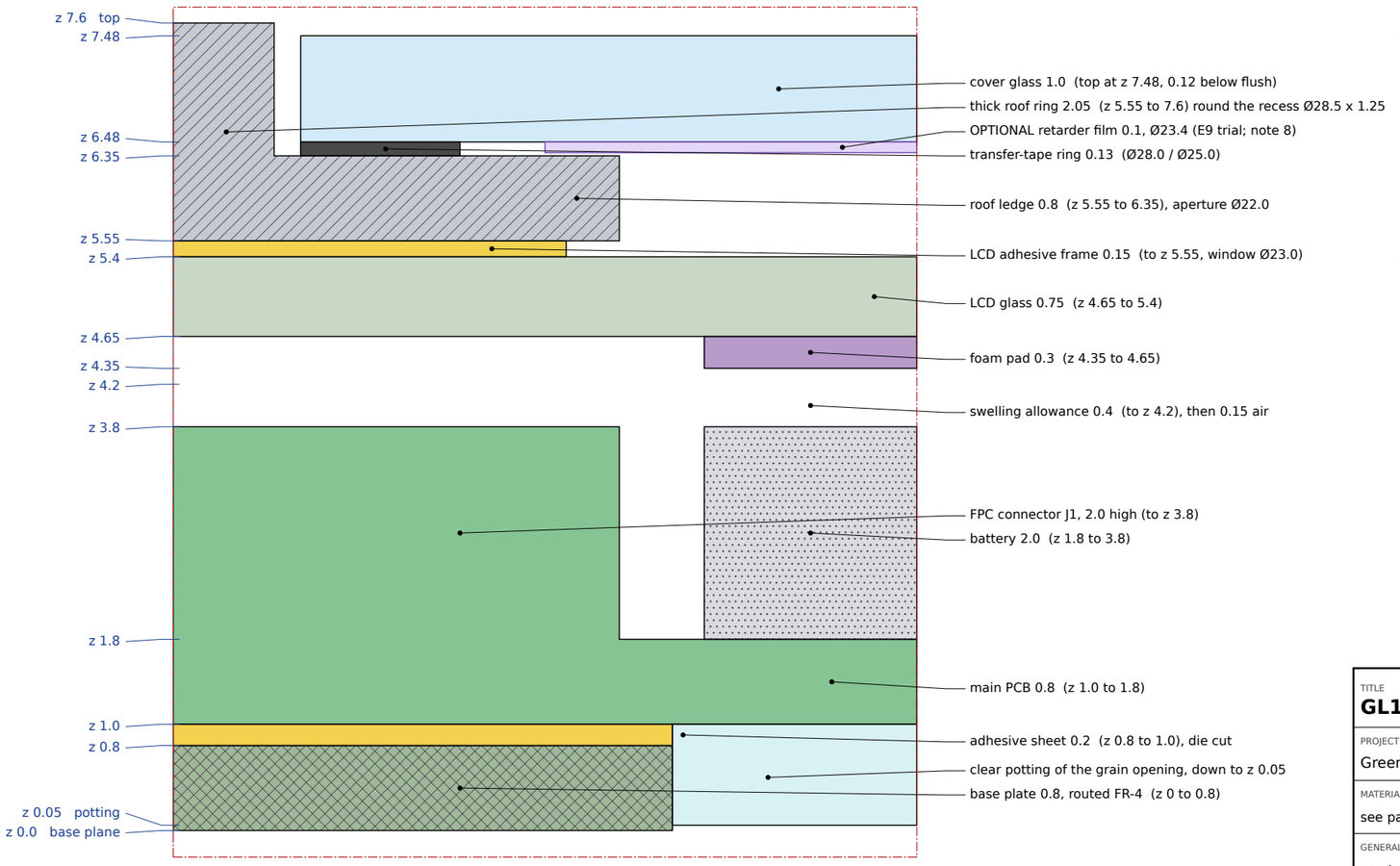
SECTION A-A on x = 0, seen from +x (visor at the right) 7:1



DETAIL C - z stack at the centre (x = 0, y = -15.2 to -8.2) 18:1



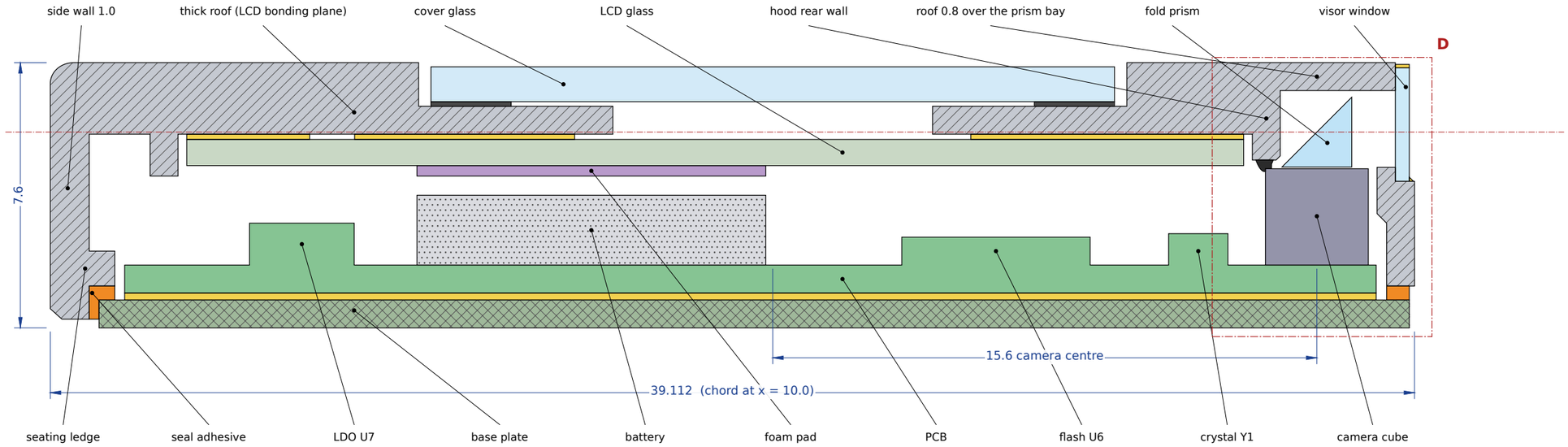
NOTES

1. Section on the plane x = 0 (the y axis through the centre). No single vertical line crosses every layer: the battery ends at y = -10.2, the ledge and tape lie outside r = 11.0. Detail C therefore spans both.
2. Rev D stack: FR-4 base plate 0.80 / adhesive sheet 0.20 / PCB 0.80 (underside at z 1.0). The posts hold the PCB top at z = 1.8; the perimeter seal joint (base top to the seal face at z 1.2) is 0.4 mm high. Alternative build (GL1-012): PA12 plate 0.87 + sheet 0.13, joint 0.33.
3. z values are heights above the base plane.
4. Purchased parts are envelope solids.
5. On x = 0 the roof is the thick roof (2.05, underside z 5.55) or the nose pad; the plain 0.8 roof (underside z 6.8) is on section B-B.
6. The tub wall ends at z = 0.25 (rim relief): base plate and ball feet are the lowest surfaces.
7. Grain opening: clear potting from the PCB down to z 0.05 at most (shown); accepted z 0.05 to 0.15. The crest of the baffle ring GL1-013 is never covered by the potting.
8. OPTIONAL (E9, EVT trial): quarter-wave film Ø23.4 x 0.1 on the glass underside, inside the Ø25.0 bore of the tape ring; 0.03 above the roof ledge outside the aperture, 0.98 above the LCD. Clocking flat 4.3 long toward the visor (y = 11.5): die cut GL1-035.

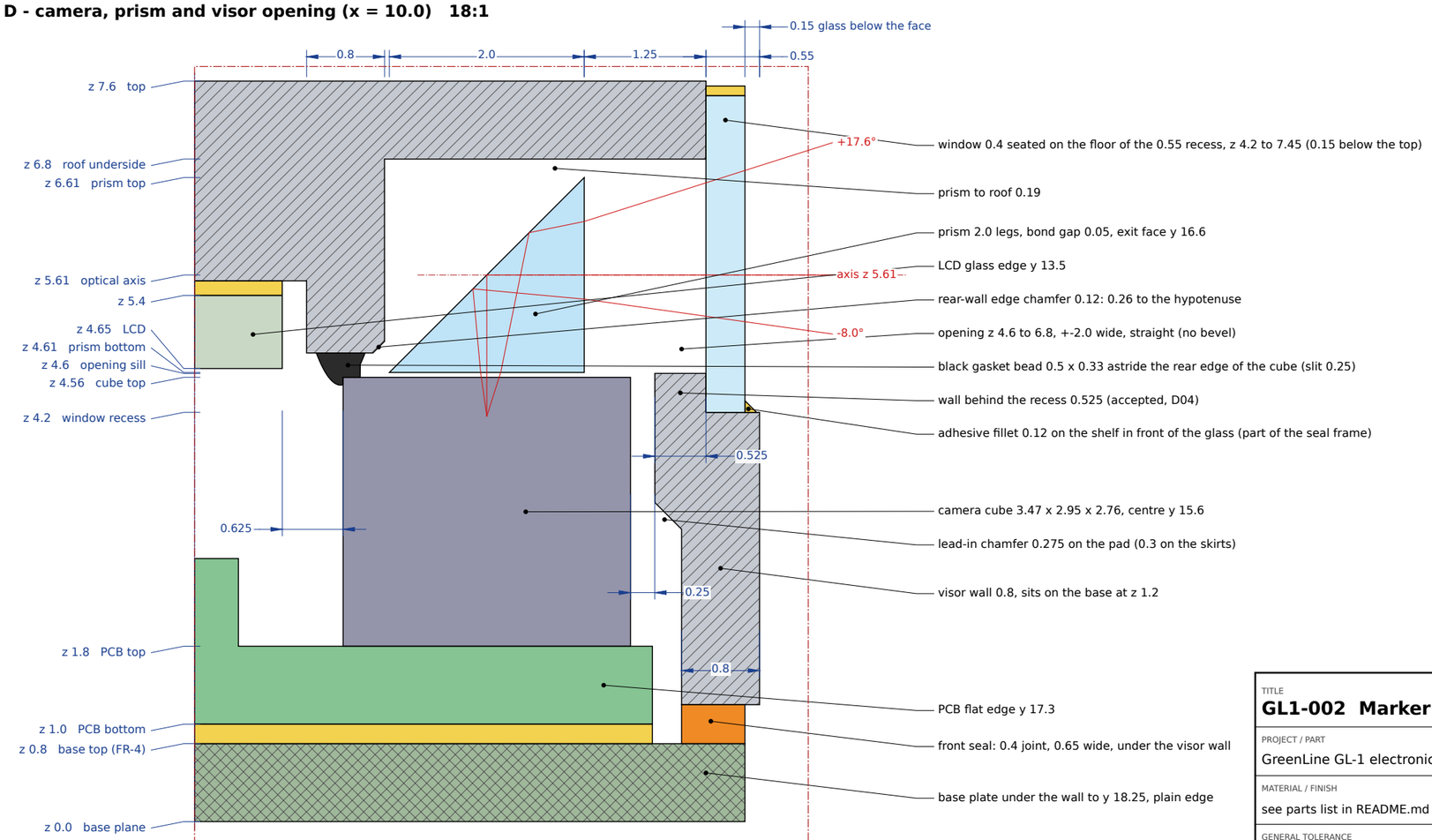
Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE		
GL1-002 Marker sections A-A and B-B		
PROJECT / PART		
GreenLine GL-1 electronic ball marker - marker assembly		
MATERIAL / FINISH	PROCESS	
see parts list in README.md	assembly (bonded)	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
nominal CAD dimensions, reference only	7:1 / 18:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	1 of 2 2026-11-13

SECTION B-B on x = 10.0 through camera R, seen from +x (visor at the right) 7:1



DETAIL D - camera, prism and visor opening (x = 10.0) 18:1



NOTES

1. Stack: base 0.8 + adhesive 0.2 + PCB 0.8 + cube 2.76 + bond 0.05 + prism 2.0 = 6.61; clearance 0.19; roof 0.8; top 7.6.
2. Red: chief rays on axis and at the vertical field edges required (+17.6° / -8.0°), traced exactly through the prism (n = 1.517). The lens pupil position is ASSUMED (optics.py).
3. The wall behind the window recess in front of the cube is 0.525 mm (accepted, D04). The 0.25-0.37 slit above the cube at the rear is closed by a black gasket bead (0.5 x 0.33, cured on the tub first; squeezed 24 % by the cube).
4. z = height above the base plane.
5. Rev D window: 5.0 x 3.25 x 0.4 borosilicate, seated on the recess floor, 0.15 below the visor face; seats 0.5 (sides) / 0.4 (sill) / 0.65 (top). Sealed by a frame of thixotropic adhesive round its edge: side gaps, top bead, sill fillet. Nothing is wicked behind the glass (YAML bond_line 0).

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE: **GL1-002 Marker sections A-A and B-B**		
PROJECT / PART: GreenLine GL-1 electronic ball marker - marker assembly		
MATERIAL / FINISH: see parts list in README.md	PROCESS: assembly (bonded)	
GENERAL TOLERANCE: nominal CAD dimensions, reference only	SCALE (A3) / UNITS: 7:1 / 18:1 mm	
STATUS: **PROVISIONAL - not released**	DWG REV / PARAMS: D-EVT1.2 / D-EVT1	SHEET / DATE: 2 of 2 2026-11-13